# Director's Pioneer Award Application

A Phase 1 LLM-Advised Robotic Whipple with Perioperative
Daraxonrasib (RMC-6236) in KRAS-Mutated PDAC

Independent Scientist | 5-Year Term | 14 Relevant Works | New Golden Age

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Intended site: UCSD Moores Cancer Center (feasibility stage)

*Submitted as an independent scientist under the funding approach set out in* Science: A New Golden Age *(July 2026). Not medical or regulatory advice, and not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. UC San Diego is named as the intended partner of choice and is not a sponsor, site, or endorser.*

## 1 The Individual Scientist and the $200 Billion Realignment

The July 2026 report to the President, *Science: A New Golden Age*, states the position this application is written against [1]. Its first overarching goal is that the United States research system should prioritize the individual scientist over legacy institutions: "If we are serious about expanding what our scientists can do, we must invest directly in American researchers and the bold ideas that drive them. Too much of our research enterprise has come to serve itself rather than the scientists within it."

Chapter II attaches a number to that goal. Federal agencies distribute approximately **$200 billion** in annual research and development funding, and the report finds the portfolio has "no systematic framework for identifying where those dollars could catalyze the greatest scientific returns, with deference instead given to the same incumbents that consume the funding." The corrective it names is to realign that portfolio toward risk taking and scale long-horizon individual specific grants "modeled on National Institutes of Health (NIH) Director's Pioneer Award."

This application takes that literally. One person, working from one public repository, has produced the package a program office produces with a team: two protocols, an Investigational New Drug application, two funding applications, five legislative drafts, and four guidance works [2], none of it carrying indirect cost at the 50 to 60 percent rates the report criticizes (Table 1).

| What the report asks for | What this application supplies |
|---|---|
| Bet on people, not just projects | One scientist with fourteen relevant, citable works in 14 months |
| Long-horizon grants of five years or more, funded in year one | A five-year term whose first year is the IND-enabling and site-agreement year |
| Gold Standard: reproducibility, transparency, negative results | Every simulation, log, and verification artifact is published with a DOI before the trial opens |

*Table 1: The report's five requests for individual scientists, and the answer this application gives to each. Sources: Science: A New Golden Age, Summary of the Report and Chapter II.*

## 2 Fit to the Director's Pioneer Award

The Pioneer Award funds a person for a long horizon rather than a project for a short one, and it is the one mechanism the report names by title, as illustrated in Figure 1.

**The unit of risk is the individual, not the aim.** The bet is that a targeted RAS(ON) inhibitor and a robotic operation can be evaluated as one governed system, with a language model confined to advisory output. It does not survive decomposition into three independent R01 aims: the drug schedule, the operative plan, and the advisory boundary are only interesting together.

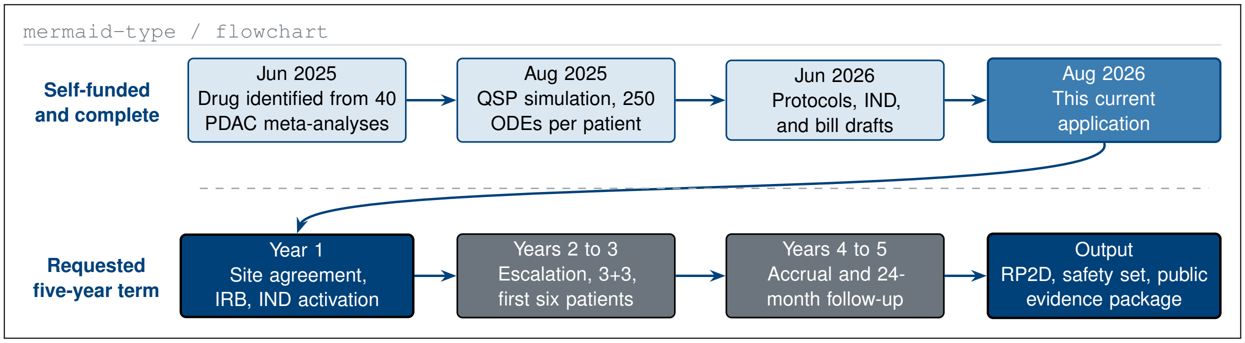


*Figure 1: Work already completed without federal funding sits above the rule; the five-year term requested here sits below it. The award enters at the one point where an individual scientist cannot proceed alone.*

**The work sits in the mid-scale gap, and the preparatory risk is retired.** Chapter II defines mid-scale science as tens of millions of dollars, teams of ten to a hundred, and half a decade, and says no institution fills it. The fourteen prior works came before any federal dollar was requested; what remains is what cannot be done alone, namely a qualified surgery program, an IRB, monitored safety infrastructure, and patients.

## 3 Evidence

In August 2025 a quantitative systems pharmacology simulation, ten arms and 250 ordinary differential equations per patient, returned a median overall survival of **12.8 months** for the daraxonrasib combination against **5.4 months** for chemotherapy [3], as shown in Table 2, Figure 2. In May 2026 the RASolute 302 readout reported **13.2** against **6.6 months** in the RAS G12 population [4]. The proximity is a chronology observation, not a validation claim: sample size was 1000 simulated against 241, intervention a combination against a single agent, and selection KRAS G12C against a primarily G12D and G12V population.

| Source | Test arm | Control | Reading |
|---|---|---|---|
| QSP simulation, Aug 2025 | 12.8 mo mOS | 5.4 mo mOS | Best arm HR 0.25; 6 of 7 patient archetypes below HR 1.00 |
| Digital twin, Oct 2025 | 12.1 mo mOS | not applicable | Best mPFS HR 0.31; credibility score 81.9 under ASME V&V 40 and ICH M15 |
| RASolute 302, May 2026 | 13.2 mo mOS | 6.6 mo mOS | Independent registrational readout in the RAS G12 population |

*Table 2: Two author simulations and one independent readout. The 2025 simulated 2.4-fold survival ratio and the 2026 observed 2.0-fold ratio are reported as a chronology, not as a validation of the simulation.*

Industry benchmarks put a single virtual trial run above $120,000 and a quantitative systems pharmacology trial above $2 million. The three simulations here cost an projected $36,330 each and took about a month each [5]: the concrete form of the report's argument that one scientist with modern tools can attempt work previously associated with larger organizations.

## 4 The Operation and Its Governance

The intervention is a staged eight-arm robotic pancreaticoduodenectomy with perioperative daraxonrasib and an on-premises large language model whose output is advisory only [6, 7]. Three boundaries make the study a governed system rather than an autonomous one.

**The model has no path to the arms.** Advisory output is rendered to a display. No generated token reaches a motion controller, and no software in the advisory path holds write access to the robot's command channel. The operating surgeon approves every motion.

d2-type / container grid

| Evidence tier | Artifact | Strength of claim | What it decided |
|---|---|---|---|
| 1. In silico | QSP, 10 arms, 250 ODEs | Hypothesis only | Arm selection |
| 2. Verified twin | 1000 patients, 55 tests | Credibility 81.9 | Sample size |
| 3. Registrational | RASolute 302 | Peer reviewed | Dose rationale |
| 4. Proposed | This Phase 1 study | No claim yet | Safety and RP2D |

*Figure 2: Four evidence tiers kept apart, so that simulation, verification, and a peer-reviewed clinical result are never read as one class. Row four is what the award would produce and carries no evidentiary weight today.*

**Stopping is bounded and measured.** Arm-level stop is specified at 3 milliseconds and system-wide stop at 500 milliseconds. Both are measured on bench before first patient and re-measured at each cohort boundary.

**Every artifact is verified before use.** The model runs on premises, pinned to a repository commit, and logs a timestamped record for each recommendation. Conversion to open operation ordered for patient safety is recorded as clinically appropriate; conversion caused by device, advisory, or integration failure is counted separately against the feasibility endpoint, as depicted in Table 3.

| Parameter | Value | Basis |
|---|---|---|
| Population | Resectable or borderline-resectable KRAS G12-mutated PDAC | Phase 1 protocol |
| Design and size | Open-label, single-arm, 3+3 dose escalation, up to 18 treated participants | Phase 1 protocol and IND |
| Co-primary endpoints | 30-day device- or procedure-related SAEs; DLT, MTD/RP2D; tasks completed with safe conversion | Phase 1 protocol |
| Follow-up | 28-day screening; 30-day, 90-day safety; 24-month OS at one qualified site, agreement pending | Phase 1 protocol |

*Table 3: Trial parameters carried unchanged from the published Phase 1 protocol and the Investigational New Drug application, so the award is evaluated against a protocol that already exists rather than one to be written later.*

**Positioning note.** *First in human refers to the integrated surgical and advisory workflow, not to daraxonrasib, which is investigational and already in Phase 3 evaluation. The robotic configuration is specified by manufacturer, model, cleared configuration, arm count, and software version at the site agreement, and no drug supply or letter of authorization is in place.*

## 5 Budget, Milestones, and the Partner Site

The request is **$700,000 per year for five years**, $3,500,000 total, no cost share [8]. The report asks that long-duration grants be fully funded in year one; with year one closes the site agreement, the IRB submission, and IND activation together rather than in sequenced as apparent in Figure 3 and Table 4.

| Year | Direct cost | Milestone | Go / no-go |
|---|---|---|---|
| 1 | $700,000 | Site agreement, IRB approval, IND activation, bench stop-latency verification | IND active and site executed |
| 2 to 3 | $1,400,000 | Dose level 1 cohort, 30-day safety review, escalation to RP2D through cohorts 2 and 3, interim advisory audit | No unexpected grade 4 event; MTD or RP2D declared |
| 4 to 5 | $1,400,000 | Full accrual to 18 participants, 90-day morbidity set, 24 mo. survival follow-up, data, code release | Accrual at plan; package released with a DOI |

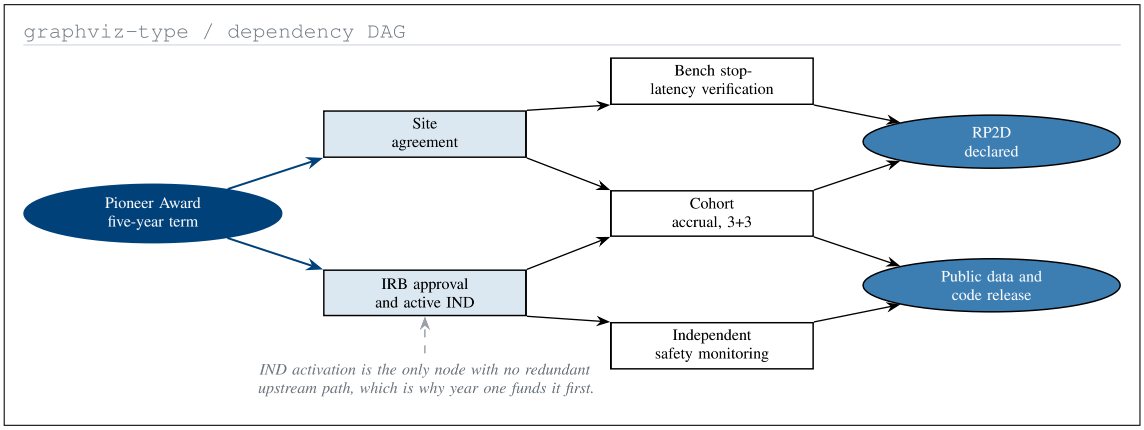


*Figure 3: What the award unblocks, and where the program still has a single point of failure. Every activity except IND activation has at least two upstream routes.*

*Table 4: Five-year milestone and go/no-go schedule at $700,000 per year. Each gate is a stated, checkable condition, so the award can be stopped at a cohort boundary rather than only at a renewal date.*

## 5.1 UC San Diego Moores Cancer Center

Moores Cancer Center is the intended partner of choice, approached at the feasibility stage only: a nonconfidential overview, a joint feasibility meeting with pancreatic surgery, gastrointestinal medical oncology, and trial operations, then an investigator-initiated intake with a budget and accrual estimate. UC San Diego is not described here as a partner, sponsor, site, or endorser, as no written authorization exists.

### Scope of Claims

Claimed: fourteen citable works, three verified simulations, and a complete regulatory package produced by one scientist in fourteen months. Not claimed: that the simulations validate RASolute 302, that the advisory system holds any clearance, that anything here has been used in a human, or that UC San Diego has agreed to anything. The trial does not exist until a site, an IRB, an active IND, monitoring, insurance, and a sponsor of record are in place.

### Availability and Conflicts

Every work cited to a DOI is openly deposited; code, verification notebooks, and figure sources are in the repository [9]. The applicant holds no equity in Revolution Medicines or any robotic surgery vendor, and no sponsor, contract research organization, or medical society reviewed this document.

## References


[1] Michael Kratsios. Science: A New Golden Age. Report to the President of the United States, July 2026. Letter of Transmittal, July 21, 2026. URL: https://www.whitehouse.gov/wp-content/uploads/2026/07/Science-A-New-Golden-Age.pdf.

[2] Kevin Kawchak. Physical AI Oncology Trial Founding Documents. ChemicalQDevice repository, July 2026. Fourteen prior works across funding, trial, and legislation. URL: https://github.com/kevinkawchak/physical-ai-oncology-trials/blob/main/funding/supplementary/Physical%20AI%20Oncology%20Trial%20Founding%20Documents.md.

[3] Kevin Kawchak. QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation: From Protocol to Prediction, Code, VVUQ, and Playbook. Zenodo, August 2025. doi:10.5281/zenodo.17001137.

[4] Eileen M. O'Reilly et al. Daraxonrasib or Chemotherapy in Previously Treated Metastatic Pancreatic Cancer. *New England Journal of Medicine*, May 2026. doi:10.1056/NEJMoa2605555.

[5] Kevin Kawchak. Daraxonrasib Efficient LLM Trial Simulations. ChemicalQDevice repository source set, July 2026. funding/supplementary/source-files/Daraxonrasib-Efficient-LLM-Trial-Simulations.zip. URL: https://github.com/kevinkawchak/physical-ai-oncology-trials/blob/main/funding/supplementary/source-files/Daraxonrasib-Efficient-LLM-Trial-Simulations.zip.

[6] Kevin Kawchak. On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma. Zenodo, June 2026. doi:10.5281/zenodo.20780121.

[7] Kevin Kawchak. Investigational New Drug Application: Daraxonrasib, Phase 1, LLM-Directed Robotic Whipple in KRAS-Mutated PDAC. Zenodo, July 2026. doi:10.5281/zenodo.21097442.

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[9] Kevin Kawchak. physical-ai-oncology-trials, repository v4.4.0. GitHub, 2026. URL: https://github.com/kevinkawchak/physical-ai-oncology-trials.